One neighbour of a binary block swap

Human Compression project: a component of the L044 recovery

Let the natural A_{2m} have a string C-marking of rank m, where $m \geq 8$ is even. Let $H = G_i$ be a transitive maximal omission preserving two blocks of size m, called its rows. Assume that every distinguished maximal omission $H_j = G_{i,j}$ is intransitive. Choose a distinguished involution l of H interchanging the rows, and suppose precisely one other mark h of H fails to commute with l. Write C for the other $m - 3$ marks. Throughout, we assume the alternating rank bound $\lfloor (q-1)/2 \rfloor$ in every smaller degree $12 \leq q < 2m$; the exact acquired finite maxima apply below degree twelve. We acquire the completed pair-block theorem under this induction, as well as the complete transitive rank theorem and the extremal symmetric independent-generator theorem used below. We also acquire the common contracted-graph reduction, whose actual-action support statements are recalled below.

Theorem 1. *If h preserves the two rows, this configuration is impossible. If h interchanges them, choosing h instead of l as the distinguished block swap gives exactly two noncommuting neighbours in H, with the same omitted label and the same binary partition. Thus the latter case belongs to the separate two-neighbour branch.*

The m orbits of l will be called pairs. Contracting them creates only a graph. The mark h need not act on its vertices as a permutation. Marks in C do act on the pairs: in coordinates (x, ϵ) matching the two rows through l, they have the form

$$(x, \epsilon) \longmapsto (x^{\pi}, \epsilon + \chi). \tag{1}$$

For the group $\langle C \rangle$ the projection to S_m is faithful. Its kernel lies in $\langle l \rangle$, and IP excludes l from $\langle C \rangle$. Consequently χ is a character of the actual projected group.

The common reduction supplies an eligible selected edge for each mark other than l. Eligibility means that its endpoints lie in distinct actual H_j-orbits. The resulting contracted forest has two components. The proved local support alternatives say that if h preserves the rows, it is a double transposition whose contracted edges are distinct, either disjoint or a path of length two. If it interchanges the rows, its contracted graph is a triangle on three pairs and it agrees with l on all other pairs. These assertions use actual matchings and evenness, not a quotient action of h.

Lemma 2. *Suppose h preserves the rows. Both its contracted edges, together with one selected edge for every mark in C, form a tree T on all m pairs. Every other contracted edge of a mark in C has both endpoints among the pairs touched by h.*

Proof. Let J consist of both h-edges. Its graph rank is two. Every selected c-edge is a bridge in the graph consisting of J and all selected C-edges. An alternative path would use no occurrence of its label; lifting that path by inserting l-edges would join its endpoints inside H_c, contradicting eligibility. Adding $m - 3$ bridges gives graph rank $m - 1$, so the graph is a connected tree on m vertices.

Let e be an additional contracted c-edge, and consider its unique path in T. That path cannot contain a selected edge of another C-label d: the new c-edge and the rest of the path would give a d-free bypass, which lifts to H_d. Thus its labels are only c, h. If its selected c-edge occurs, neither end of the path can be on that edge, since c induces an involution on the pairs

and its distinct transposition edges are disjoint. The first and last path edges must therefore both be the two h-edges; the path is h, c, h. Otherwise the path uses only h. In either case both endpoints of e belong to the set Y of pairs touched by h. Flips within one pair are omitted from this contracted graph and remain controlled by (1). □

Every mark of C except a possible neighbour c of h centralizes both l, h. If the two contracted h-edges are disjoint, the joint group has two four-point orbits over Y; if they meet, it has one six-point orbit there. All its other orbits have size two. A centralizer preserves this union of the larger orbits, so these marks preserve Y. Transitivity and Lemma 2 imply that c exists and its selected edge is the unique edge of T leaving Y. There are no additional edges leaving Y by that lemma.

Lemma 3. *The two contracted h-edges cannot be disjoint.*

Proof. Here Y consists of four vertices. The unique exit from Y forces the two h-components to be joined inside Y in T. There is exactly one joining selected edge, labelled $g \in C$, and the inside path has labels h, g, h. The mark g is different from c, whose selected edge leaves Y. Since g centralizes l, h, it interchanges the two four-point joint orbits, and therefore moves every vertex of Y in projection. In particular it does not commute with c: the latter sends the attachment vertex outside Y, whereas g sends it to another vertex of Y. The two compositions disagree because c has only one exit.

There are $m - 4 \geq 4$ outside vertices. Their tree is connected, so an edge d is incident with the outside endpoint of c. It also fails to commute with c. In projection, d sends that outside endpoint to a different outside vertex, whereas its action on the inside endpoint remains within Y; applying c there cannot reach the new outside vertex. Thus h, g, d are three distinct noncommuting neighbours of c, impossible in a string. □

A path on three pairs

Name the three pairs a, u, b, and choose their row coordinates so that

$$h = (a_0\ u_0)(u_1\ b_1). \tag{2}$$

The graph of l, h on their six points is the path

$$a_1, a_0, u_0, u_1, b_1, b_0.$$

Every mark $g \in C \setminus \{c\}$ restricts to an automorphism of this labelled path, hence to its identity or reversal. In projection on Y these are respectively the identity and $\tau = (a\ b)$; their row characters are respectively zero and one. The selected c-edge joins a vertex of Y to an outside vertex x. The outside tree is connected.

Lemma 4. *If that vertex is the middle pair u, the completed pair-block branch excludes the configuration.*

Proof. Put $z' = (lh)^3$. On the six-point path this is reversal, pairing a_0 with b_1, a_1 with b_0, and u_0 with u_1. Outside it, $z' = l$. Thus z' is fixed-point-free, belongs to H, interchanges the binary rows and centralizes l, h. All marks other than c centralize it. In projection c has the edge $(u\ x)$ and possibly $(a\ b)$, its only possible extra edge by matching and Lemma 2. Either permutation preserves the pairs of z', with either row character, and commutes with z'. Hence z' is central in H.

We justify maximality of this new pair system. In coordinates matching the rows through z', any permutation preserving their binary partition and centralizing z' lies in $\mathrm{diag}(S_m) \times \langle z' \rangle$. Let P be the transitive image of H on the new pairs. If all $m - 1$ projected marks are independent,

Whiston makes $P = S_m$. Otherwise choose a redundant projected mark j. The actual H_j still projects onto P, but is intransitive on the $2m$ points. It has two orbits, each meeting every new pair once, so its projection is faithful. These two orbits are sections of the new pairs, and need not be the original binary rows. It supplies a genuine transitive string C-marking of P of rank $m-2$. The complete acquired L047 theorem makes $P = S_m$ or A_m: for $m \geq 10$, $m-2 > m/2+1$, and the complete degree-eight proper transitive boundary has no rank-six case. Thus the new pair system is maximal. The completed pair-block theorem for the parent A_{2m} now excludes it, with its stated induction hypotheses. □

We may therefore attach at an outer pair, say a, so that the selected edge is $(a\ x)$. A mark $g \neq c$ with extra edge $\tau = (a\ b)$ cannot commute with c, by its unique exit from Y. Since c already has h as a noncommuting neighbour, at most one such g exists. The selected edge at the outside endpoint x also fails to commute with c, so that possible g must label that first outside edge. The outside tree has degree at most two: three incident edges would give three pairwise noncommuting projected transpositions and a triangle in the string diagram. Its attachment vertex x has degree one, because two incident outside edges would give c two noncommuting neighbours in addition to h. Thus this tree is a path starting at x. Write its first edges as $(x\ y), (y\ z), \ldots$. There is a second edge because the outside set has $m - 3 \geq 5$ vertices.

Lemma 5. *No $g \neq c$ has the extra edge $(a\ b)$.*

Proof. If it did, its projected mark would be $(a\ b)(x\ y)$. The projection of c could not also have $(b\ u)$: then the whole projected C graph would be connected, making $H_h = \langle l, C\rangle$ transitive, contrary to hypothesis. Thus the projected first marks are

$$(a\ x), \quad (a\ b)(x\ y), \quad (y\ z), \ldots.$$

They generate the natural S_{m-1} fixing u. The first two form a transitive dihedral core on four indices; conjugating the first attached transposition by that core supplies all star transpositions on five indices. Subsequent path transpositions adjoin one index at a time. The second tail mark $(y\ z)$ has row character zero, since its restriction to the six-point path is the identity. A character of S_{m-1} vanishing on a transposition is trivial, as transpositions are conjugate and generate the group. But the character of g is one, because it reverses the six-point path. This contradiction proves the claim. □

All marks of C except c are consequently ordinary doubled path transpositions with character zero. The projection of c is $(a\ x)$, possibly multiplied by $(b\ u)$. If the extra factor is absent, the projected group is the natural S_{m-2} on a and the outside indices. Its character is trivial, since one of its transposition generators has character zero. This is the canonical case treated below. We next exclude the extra factor.

Lemma 6. *Let $\bar{c}$ be the doubled permutation $(a\ x)(b\ u)$ and let d be the doubled transposition $(x\ y)$. Neither $c = \bar{c}$ nor $c = \bar{c}l$ is compatible with the intersection property of the actual marks l, h, c, d.*

Proof. Write c_0, c_1 for the restrictions of $\bar{c}$ to the two rows. First take $c = \bar{c}$. The element $v = (hc)^2$ is the pure row-zero double transposition $(a\ u)(b\ x)$. Its conjugate by lhl is the different double transposition $(a\ b)(u\ x)$ on that row, and their product is c_0. Thus $\langle l, h, c\rangle$ contains c_0 and $c_1 = cc_0$.

On the other hand, $e = (cd)^3$ is the doubled $(b\ u)$, so he is a pure row-zero three-cycle on $\{a, b, u\}$. It and its c-conjugate generate A_4 on $\{a, b, u, x\}$. Indeed they are even, their supports make their action transitive on four points, and their generated order is divisible by four and three; it must equal twelve. Hence $\langle h, c, d\rangle$ also contains c_0 and c_1. But $\langle h, c\rangle$ is dihedral of order

eight, with a faithful action on row zero: there its two generators have product of order four. It cannot contain the nonidentity c_1 supported only on row one. This contradicts

$$\langle l, h, c\rangle \cap \langle h, c, d\rangle = \langle h, c\rangle.$$

Now take $c = \bar{c}l$. This time $e = (cd)^3$ is the doubled $(b\ u)$ multiplied by l. Put $q = (he)^2$. It is the diagonal copy of the preceding three-cycle on $\{a, b, u\}$. The elements q, q^c generate diagonal A_4 on $\{a, b, u, x\}$. Together with d they generate diagonal S_5 on those four indices and y: conjugating d by the transitive A_4 gives all star transpositions to y. In particular $\langle h, c, d\rangle$ contains $\bar{c}$, and hence contains $l = \bar{c}c$. This contradicts IP between the singleton mark l and the disjoint marked set $\{h, c, d\}$. Both exclusions are uniform permutation identities; no finite enumeration is used as a premise. □

The canonical case and the missing generator

Only the following actual marking of H remains: l flips all pairs, h is (2), and C is the doubled Coxeter path generating S_O, where O consists of a and all the outside indices. Its noncommutation diagram is the connected path $l, h, c, d, \ldots$. Therefore the omitted label i is an endpoint of the full string; removing an interior label would leave two nonempty commuting components. There are two orientations for the omitted even involution σ.

If it is before l, it centralizes $J = \langle h, C\rangle$. The actual J-orbits have sizes $m-1, m-2, 2, 1$, on

$$O_0 \cup \{u_0\}, \qquad O_1, \qquad \{u_1, b_1\}, \qquad \{b_0\}.$$

These sizes are distinct. The induced groups on the first two orbits are the full natural S_{m-1}, S_{m-2}, whose centralizers are trivial. Thus σ fixes those orbits pointwise and fixes the singleton. Its only possible nontrivial action is the transposition $(u_1\ b_1)$, which is odd. No nonidentity even σ is possible.

If it is after the tail, it centralizes K generated by l, h and all but the last mark of C. Let t be the final index. The actual K-orbits are the pair over t and its complement of size $2(m-1)$. Transitivity on that complement follows from the connected contracted path and its l-edges. On it, the remaining C marks generate diagonal $S_{O'}$ with $O' = O \setminus \{t\}$. Conjugating h by this group gives

$$h_x = (x_0\ u_0)(u_1\ b_1) \quad (x \in O').$$

The products $h_x h_y$ generate the pure alternating group on $O'_0 \cup \{u_0\}$, of size $m-2$. A permutation centralizing K preserves this support and centralizes its natural alternating action, whose centralizer is trivial. It fixes a point of the large K-orbit, hence fixes that whole orbit by transitivity. Thus σ can only flip the isolated pair over t, again an odd transposition. This excludes the second orientation and proves the row-preserving part of Theorem 1.

Finally, if h interchanges the binary rows, the acquired support lemma gives one six-point orbit of $\langle l, h\rangle$ and all other orbits of size two. If every mark in C commuted with h, it would centralize this group and preserve that unique six-point orbit, contradicting transitivity of H. Hence another neighbour c of h belongs to C. String relations allow no others. Choosing h as the distinguished block swap therefore makes its noncommuting remainder precisely $\{l, c\}$. This is the exact same-partition reduction to the two-neighbour case asserted by the theorem.

Attribution and scope. The target is the one-neighbour part of source Proposition 3.5, whose graph strategy and displayed canonical path remain attributed to Cameron–Fernandes–Leemans–Mixer. The safe contracted support and bridge arguments were developed by the root investigator; the present continuation supplies the alternative-pair maximality, character obstruction, explicit local IP contradictions and actual endpoint centralizers. The row-swapping case remains a reduction to the separate two-neighbour case. No whole-source recovery is claimed here.